Enterprise Network Cyber Attack & Intrusion Detection using NSL-KDD

Machine Learning-Based Intrusion Detection System

Models: Decision Tree (Gini), Decision Tree (Entropy), Gaussian Naive Bayes, LinearSVC, Balanced LinearSVC

Dataset: NSL-KDD — KDDTrain+ and KDDTest+

Classification: Normal, DoS, Probe, R2L, U2R

Final Project Report

Academic Machine Learning & Cybersecurity Project

Random state: 42

Abstract

This project develops and evaluates a machine-learning-based Enterprise Network Intrusion Detection System (IDS) using the NSL-KDD benchmark dataset. The task is formulated as five-class classification covering Normal traffic and four attack categories: Denial of Service (DoS), Probe, Remote-to-Local (R2L), and User-to-Root (U2R). Five classical machine-learning models are compared: Decision Tree using Gini impurity, Decision Tree using entropy, Gaussian Naive Bayes, LinearSVC, and class-balanced LinearSVC.

The models are evaluated on the official KDDTest+ test set using accuracy, macro precision, macro recall, macro F1-score, per-class metrics, five-fold stratified cross-validation, generalization gap, and prediction latency for 10,000 connection records. The entropy Decision Tree achieved the strongest overall test performance with 77.56% accuracy and 56.14% macro F1, while also providing the lowest measured prediction latency at 2.428 ms per 10,000 records.

The balanced LinearSVC improved macro F1 from 47.03% to 51.76% relative to the baseline LinearSVC and substantially improved recall for the minority R2L and U2R classes. The experiments also reveal substantial differences between cross-validation and KDDTest+ performance, highlighting the challenge of generalization across network attack distributions.

Keywords: Intrusion Detection System, NSL-KDD, Decision Tree, Gaussian Naive Bayes, LinearSVC, class imbalance, macro F1, network security, machine learning.

1. Introduction

Enterprise networks generate large volumes of connection records containing information about protocols, services, connection flags, bytes, durations, error indicators, and other traffic characteristics. Detecting malicious activity within these records is an important cybersecurity task because attacks can resemble legitimate traffic and different attack families can have substantially different characteristics.

An Intrusion Detection System attempts to identify suspicious or malicious network activity. In a machine-learning IDS, a classifier learns patterns from labelled historical traffic and assigns a class to previously unseen connection records.

This project evaluates classical supervised learning methods using NSL-KDD and focuses on five traffic categories: Normal, DoS, Probe, R2L, and U2R. Because the dataset is strongly imbalanced, the study emphasizes macro-averaged metrics and minority-class recall rather than relying on accuracy alone.

1.1 Problem Statement

Develop and evaluate a machine-learning-based IDS capable of distinguishing normal network connections from DoS, Probe, R2L, and U2R attacks using the NSL-KDD dataset, while examining class imbalance, generalization to the official KDDTest+ benchmark, and prediction efficiency.

1.2 Objectives

- Build a consistent five-class IDS classification pipeline using NSL-KDD.
- Evaluate Decision Tree, Gaussian Naive Bayes, and LinearSVC-based classifiers.
- Compare baseline and class-balanced LinearSVC behaviour.
- Use macro-averaged metrics so minority classes influence evaluation.
- Measure cross-validation performance and generalization to KDDTest+.
- Measure prediction latency for a fixed batch of 10,000 connection records.

2. Dataset

The NSL-KDD dataset provides labelled network connection records derived from the KDD intrusion detection benchmark. The project uses KDDTrain+ for training and the official KDDTest+ split as the primary unseen test benchmark.

2.1 Dataset Size

Split	Records	Final Features
KDDTrain+	125,973	122
KDDTest+	22,544	122

The original dataset contains 43 columns. After preprocessing, the final model representation contains 122 numeric features.

2.2 Target Classes

Class	Description
Normal	Legitimate network connections
DoS	Denial-of-Service attack traffic
Probe	Scanning and reconnaissance activity

Class	Description
R2L	Remote-to-Local attack attempts
U2R	User-to-Root privilege escalation attacks

2.3 Class Imbalance

The dataset is strongly imbalanced. Normal and DoS dominate the test set, while U2R contains only 67 records. Therefore, accuracy alone can hide poor performance on rare attack families. Macro precision, macro recall, macro F1, and per-class recall are used to provide a more balanced evaluation.

3. Data Preprocessing

The preprocessing pipeline transforms the original mixed-type NSL-KDD records into a 122-dimensional numeric representation.

3.1 Categorical Encoding

The categorical features protocol_type, service, and flag are transformed using one-hot encoding with unknown categories ignored.

3.2 Scaling

Continuous features are transformed using RobustScaler. Six binary features are retained without scaling. The difficulty attribute is excluded from the model representation.

3.3 Final Representation

- Scaled continuous features.
- Six unscaled binary features.
- One-hot encoded categorical features.
- Final dimensionality: 122 features.

The processed arrays were verified to contain no NaN values or infinite values. X_train has shape (125973, 122) and X_test has shape (22544, 122).

4. Machine Learning Models

4.1 Decision Tree

Decision Trees recursively partition the feature space using feature thresholds. Two variants were evaluated: Gini impurity and entropy as the splitting criterion. Both models use random_state=42.

4.2 Gaussian Naive Bayes

Gaussian Naive Bayes models each feature using a class-conditional Gaussian distribution and combines feature likelihoods with class priors. Its conditional independence assumption provides a simple probabilistic baseline.

4.3 LinearSVC

LinearSVC learns linear decision boundaries between classes. The final configuration uses C=0.1, dual=False, tol=1e-3, max_iter=20,000, and random_state=42.

Implementation limitation: convergence warnings were observed for LinearSVC during the experiments. Therefore, the results should not be interpreted as proof of complete optimization convergence.

4.4 Balanced LinearSVC

Balanced LinearSVC uses class_weight='balanced' to increase the relative importance of minority classes during model training.

5. Evaluation Methodology

5.1 Primary Test Evaluation

The official KDDTest+ split is treated as the primary unseen benchmark. Accuracy, macro precision, macro recall, macro F1, and per-class precision, recall, and F1 are measured.

5.2 Five-Fold Cross-Validation

Five-fold StratifiedKFold is performed with shuffle=True and random_state=42. The reported values are the mean and standard deviation over five folds.

CV limitation: the preprocessing representation was fitted on the complete training data before cross-validation. Therefore, the CV experiment evaluates the models on a fixed training-derived representation rather than performing fully fold-wise preprocessing. This is reported transparently as a methodological limitation.

5.3 Generalization Gap

The generalization gap is calculated as the difference between CV macro F1 and KDDTest+ macro F1. A large positive gap indicates that performance observed within the training distribution does not transfer equally well to the official test distribution.

5.4 Prediction Latency

Prediction latency is measured using 10,000 records from X_test. A 100-record warm-up is performed, followed by five prediction repetitions. Training time is excluded from the benchmark.

6. Results

6.1 Overall Test Results

Model	Accuracy	Macro F1	Latency / 10k
DT — Entropy	77.56%	56.14%	2.428 ms
DT — Gini	75.86%	55.46%	2.983 ms
Balanced LinearSVC	73.19%	51.76%	8.919 ms
LinearSVC	71.67%	47.03%	9.430 ms
Gaussian NB	33.23%	24.89%	48.162 ms

The entropy Decision Tree achieved the strongest overall test performance. Balanced LinearSVC was the strongest linear model by macro F1 and macro recall.

6.2 Minority Attack Detection

Minority classes remain difficult. The entropy Decision Tree achieved only 8.35% R2L recall and 13.43% U2R recall. Gaussian Naive Bayes achieved 64.18% U2R recall, but its U2R precision was only 1.06%, resulting in an F1-score of 2.08%. This shows why recall must be interpreted together with precision.

6.3 Effect of Class Balancing

Relative to baseline LinearSVC, class weighting increased accuracy from 71.67% to 73.19%, macro recall from 46.80% to 55.64%, and macro F1 from 47.03% to 51.76%. R2L recall increased from 0.66% to 12.86%, while U2R recall increased from 2.99% to 32.84%.

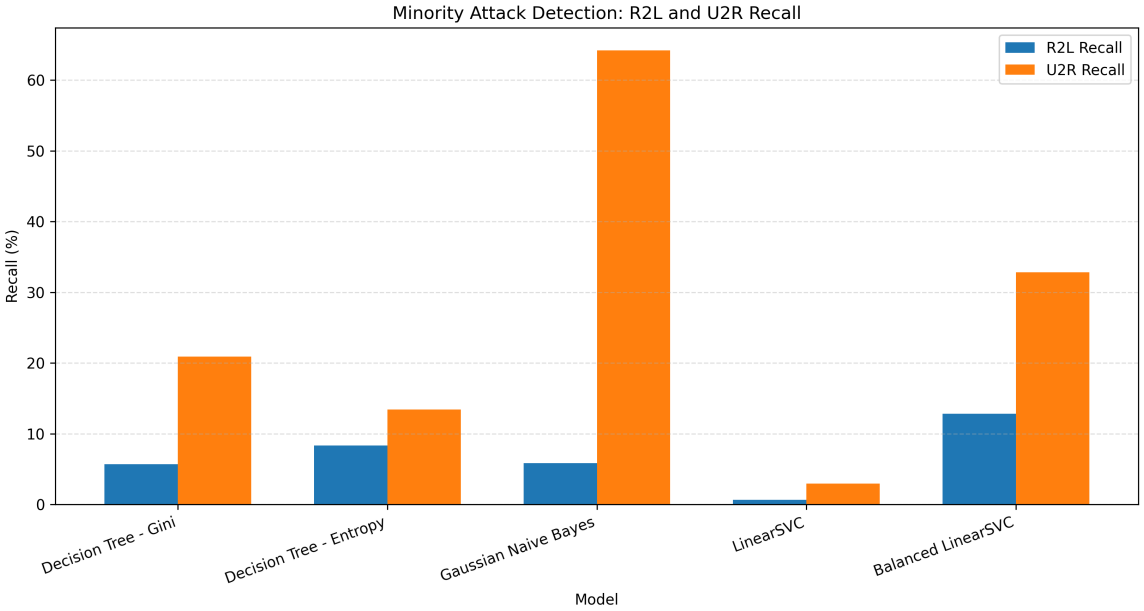


Figure 1. Minority attack recall comparison.

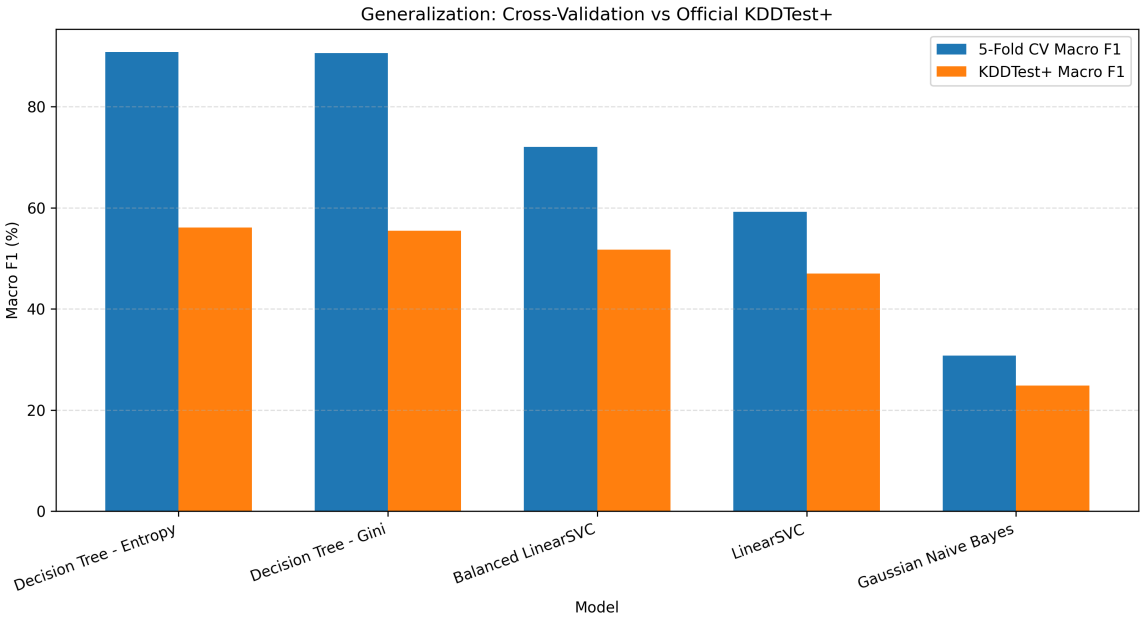


Figure 2. Cross-validation to KDDTest+ macro F1 generalization gap.

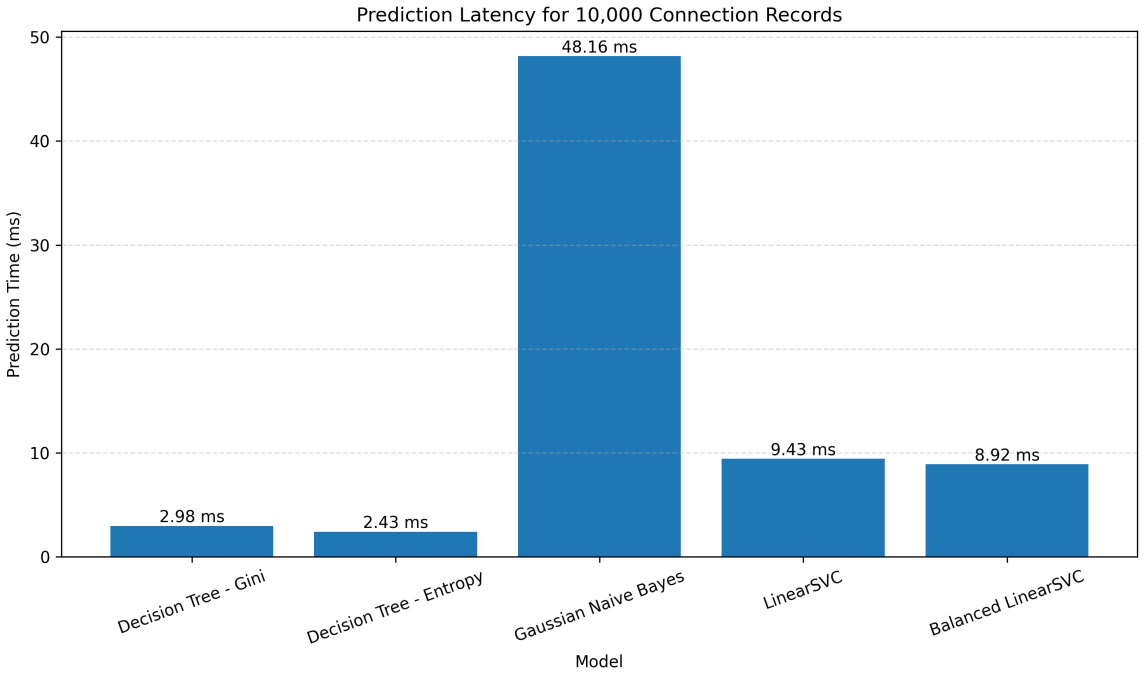


Figure 3. Prediction latency for 10,000 connection records.

6.4 Confusion Matrices

The confusion matrices provide class-level insight into prediction errors and highlight the difficulty of identifying R2L and U2R traffic.

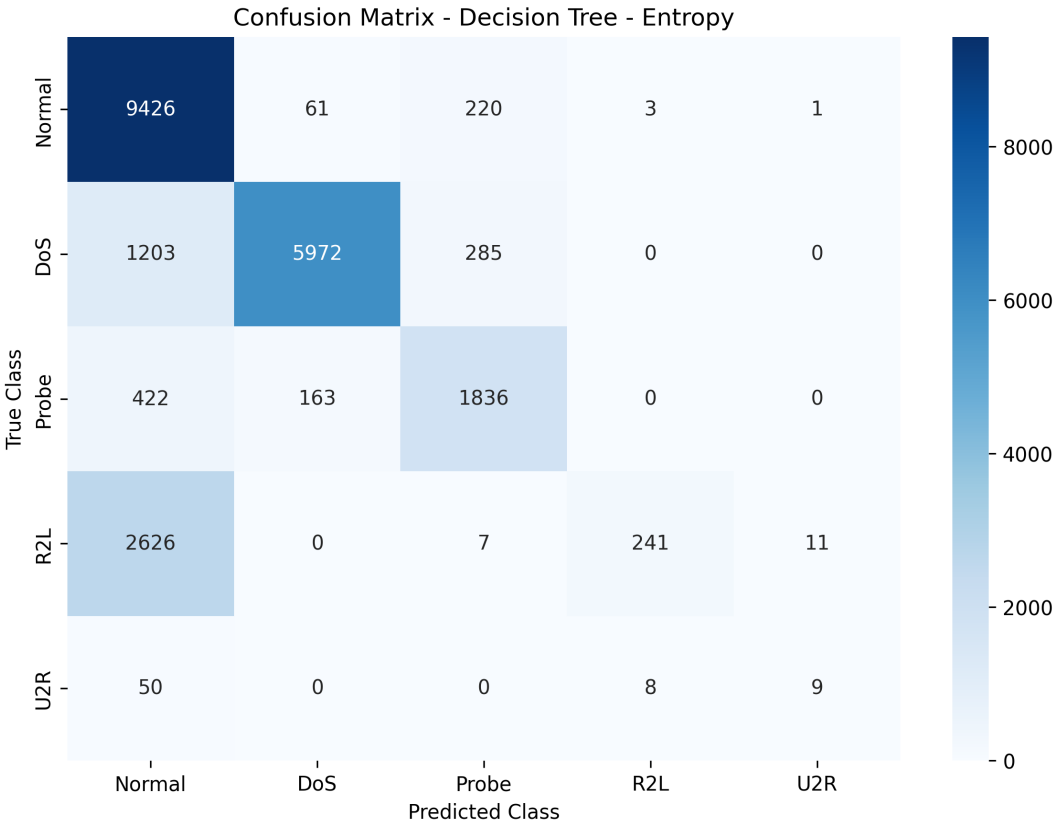


Figure 4. Decision Tree — Entropy confusion matrix.

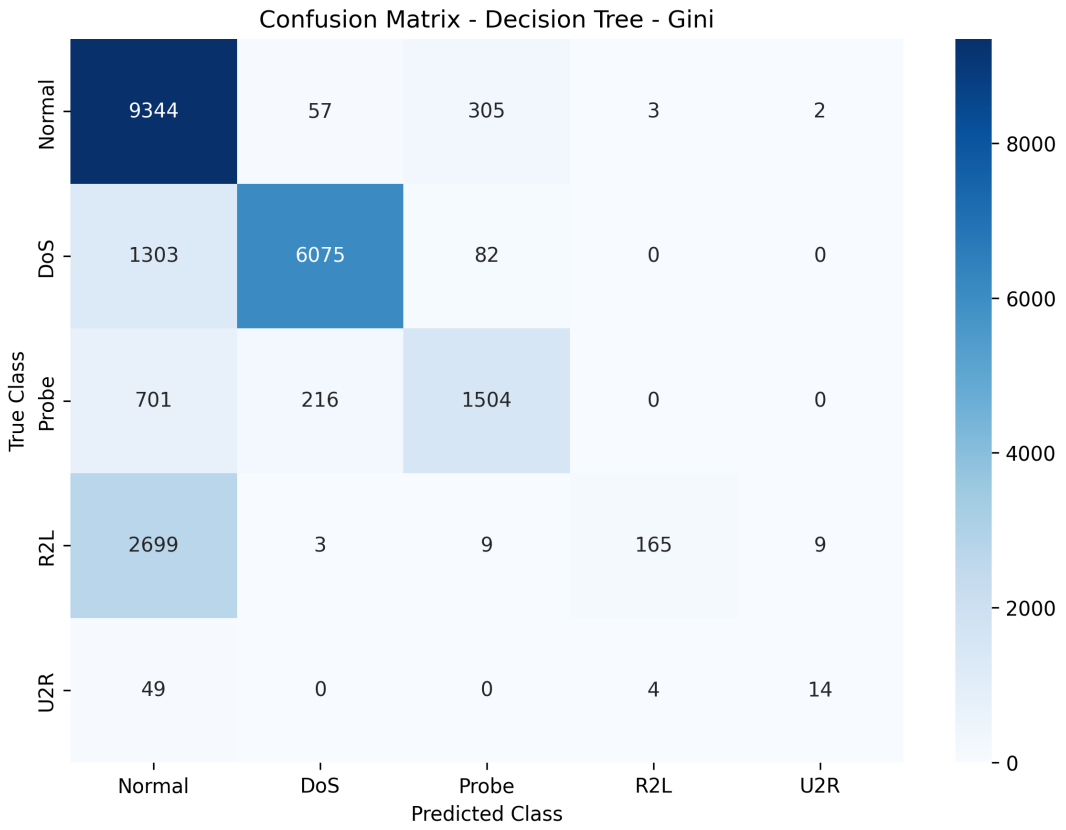


Figure 5. Decision Tree — Gini confusion matrix.

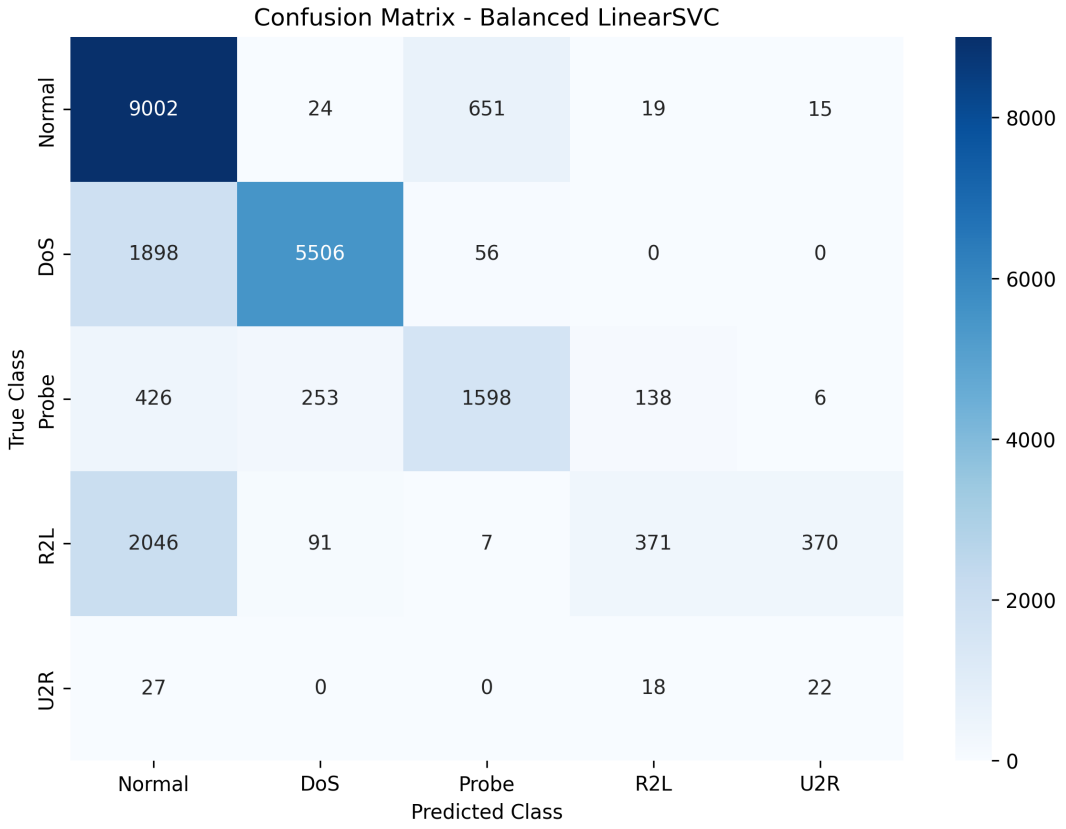


Figure 6. Balanced LinearSVC confusion matrix.

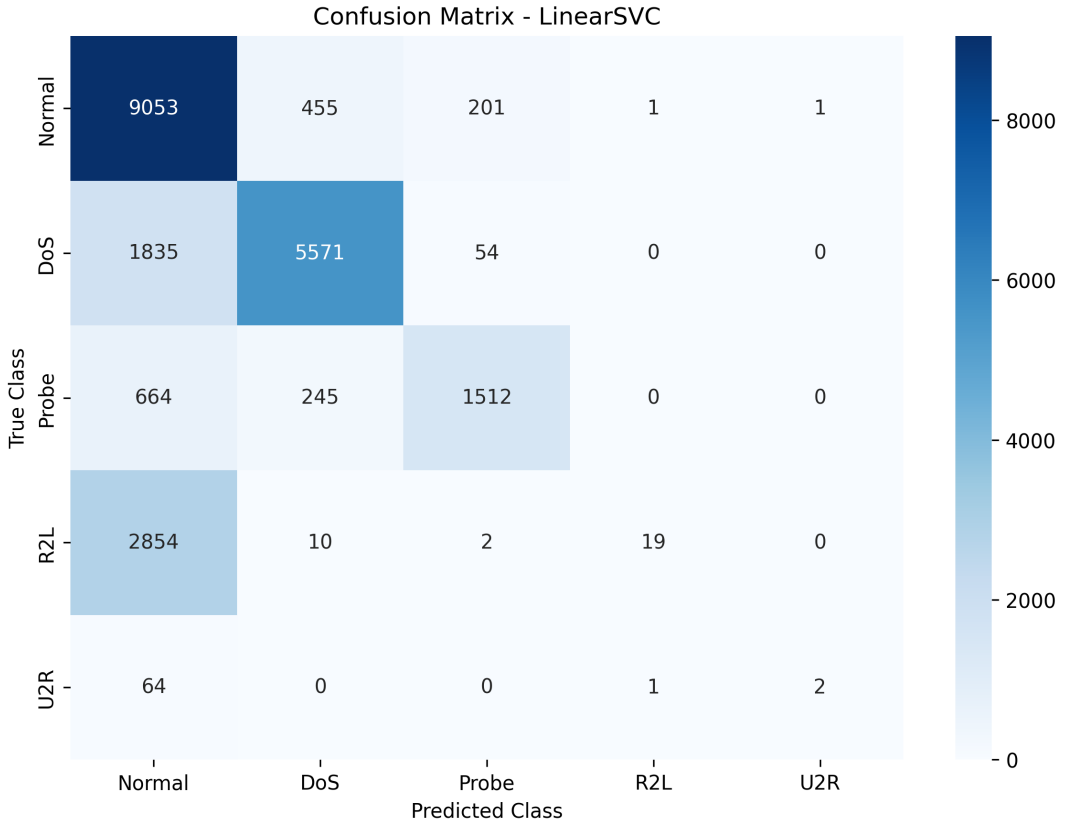


Figure 7. LinearSVC confusion matrix.

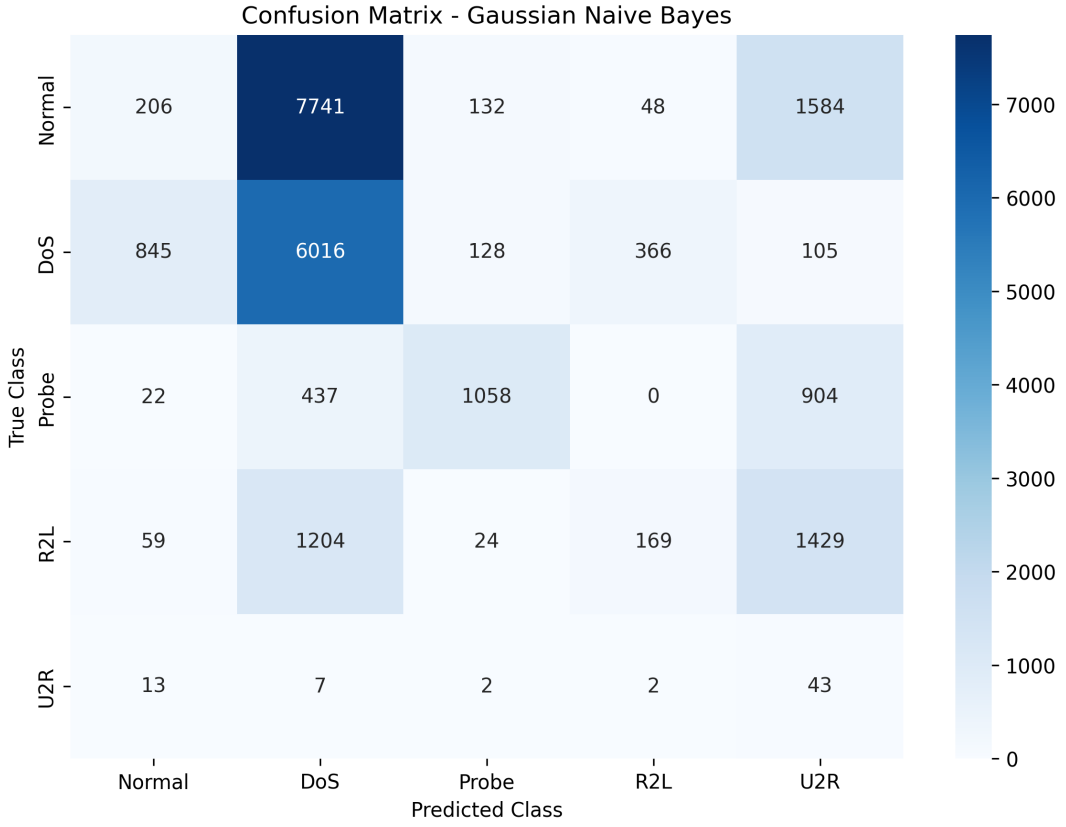


Figure 8. Gaussian Naive Bayes confusion matrix.

6.5 Five-Fold Cross-Validation

Model	Accuracy	Precision	Recall	Macro F1
DT Entropy	99.802%	91.893%	90.469%	90.789%
DT Gini	99.781%	90.021%	91.480%	90.575%
Balanced LinearSVC	94.694%	66.860%	87.547%	72.054%
LinearSVC	96.566%	61.135%	58.568%	59.183%
Gaussian NB	41.111%	42.168%	53.847%	30.777%

The CV ranking is DT Entropy, DT Gini, Balanced LinearSVC, LinearSVC, and Gaussian Naive Bayes. The official KDDTest+ results remain the primary benchmark for unseen-data performance.

6.6 Generalization Gap

Model	CV F1	Test F1	Gap
DT Entropy	90.7894%	56.14%	34.6494 pp
DT Gini	90.5747%	55.46%	35.1147 pp
Balanced LinearSVC	72.0541%	51.76%	20.2941 pp
LinearSVC	59.1834%	47.03%	12.1534 pp
Gaussian NB	30.7766%	24.89%	5.8866 pp

The large Decision Tree gaps indicate that strong performance within the training-derived distribution does not transfer proportionally to KDDTest+. This is consistent with distribution differences and attack subtype variation. The gap should not be described solely as conventional overfitting.

6.7 Prediction Latency

Model	Average Time / 10k	Connections / ms
DT Entropy	2.428 ms	4118.94
DT Gini	2.983 ms	3351.85
Balanced LinearSVC	8.919 ms	1121.19
LinearSVC	9.430 ms	1060.46
Gaussian NB	48.162 ms	207.63

7. Discussion

7.1 Overall Model Comparison

The entropy Decision Tree provides the best overall balance between detection quality and prediction speed in this experiment. Its 56.14% test macro F1 is the highest among the five evaluated models.

7.2 Class Imbalance

The class-balanced LinearSVC experiment demonstrates that changing the training objective can improve minority recall. However, the improved recall comes with lower precision, particularly for U2R. In a real IDS, the acceptable trade-off depends on the relative cost of missed attacks and false alarms.

7.3 Generalization

The CV-to-test gap is one of the most important findings. Decision Trees show very high CV macro F1 but substantially lower KDDTest+ macro F1. This suggests that within-training validation alone can give an overly optimistic view of IDS performance when the test distribution differs.

7.4 Gaussian Naive Bayes

Gaussian Naive Bayes performs substantially worse overall, with 33.23% accuracy and 24.89% macro F1 on KDDTest+. The conditional-independence and Gaussian assumptions appear poorly matched to the resulting network feature representation.

7.5 Trees versus Linear Models

The Decision Trees outperform the linear classifiers on test macro F1, suggesting that nonlinear feature interactions and threshold-based relationships are useful for this classification task. LinearSVC remains valuable as a comparatively simple baseline and becomes more effective for minority recall when class weighting is used.

7.6 Operational Efficiency

The entropy Decision Tree is the fastest measured model at 2.428 ms per 10,000 connection records. Gaussian Naive Bayes is the slowest at 48.162 ms under the same experimental benchmark.

8. Limitations

- NSL-KDD is a benchmark dataset and may not fully represent modern enterprise traffic.
- R2L and U2R have very limited representation, making robust learning difficult.
- Cross-validation used a preprocessing representation fitted before fold splitting.
- LinearSVC produced convergence warnings under the evaluated configuration.
- Latency measurements are environment-specific and are not a production throughput guarantee.
- Only classical supervised models were evaluated.
- No live network or streaming deployment was implemented.

9. Future Work

- Evaluate SMOTE and other imbalance-handling methods without information leakage.
- Evaluate ensemble methods such as Random Forest and gradient-boosted classifiers.
- Perform explicit drift analysis between training and test distributions.
- Investigate feature engineering and feature selection for minority attacks.
- Tune decision thresholds using security-oriented false-positive and false-negative costs.
- Evaluate additional and more recent intrusion-detection datasets.
- Integrate the classifier into a streaming or near-real-time IDS pipeline.
- Add explainability methods to help security analysts interpret predictions.

10. Conclusion

This project demonstrates a complete classical machine-learning workflow for five-class enterprise network intrusion detection using NSL-KDD. Five models were implemented and evaluated using the official KDDTest+ benchmark, macro-averaged metrics, stratified cross-validation, generalization analysis, and prediction latency.

The Decision Tree using entropy achieved the strongest overall test result, with 77.56% accuracy and 56.14% macro F1, while also providing the lowest measured prediction latency. The class-balanced LinearSVC provided a complementary strength by improving minority-class recall and macro F1 relative to the baseline LinearSVC.

The experiments demonstrate that IDS quality cannot be represented by accuracy alone. The strong disparity between common and rare attack classes and the substantial cross-validation-to-test gaps emphasize the importance of minority-class evaluation and testing on an unseen distribution.

11. Reproducibility

- Environment: Google Colab.
- Random state: 42.
- Training records: 125,973.
- Official test records: 22,544.
- Final feature count: 122.
- Primary notebook: Nami_NB_SVM_IDS.ipynb.
- Decision Tree notebook: Alma_DT_IDS.ipynb.
- Raw datasets and processed serialized data are excluded from GitHub.
- Evaluation figures and CSV summaries are included in the repository.

12. References

[1] M. Tavallaee, E. Bagheri, W. Lu, and A. A. Ghorbani, "A Detailed Analysis of the KDD CUP 99 Data Set," Proceedings of the Second IEEE Symposium on Computational Intelligence for Security and Defence Applications, 2009.

[2] M. Tavallaee, E. Bagheri, W. Lu, and A. A. Ghorbani, "NSL-KDD dataset," Canadian Institute for Cybersecurity / benchmark distribution.

[3] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[4] Scikit-learn documentation, including DecisionTreeClassifier, GaussianNB, and LinearSVC API references.

Appendix A — Final Project Summary

Best overall model: Decision Tree — Entropy

Best test accuracy: 77.56%

Best test macro F1: 56.14%

Fastest prediction: Decision Tree — Entropy, 2.428 ms / 10,000 records

Best minority-focused linear model: Balanced LinearSVC

Highest U2R recall: Gaussian Naive Bayes, 64.18%, with 1.06% precision

Primary benchmark: Official KDDTest+

Final feature dimensionality: 122